WORKED SOLUTION

STEP III 1987, Question 11

Mechanics • Relative Velocity Optimisation

STEP III 1987, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

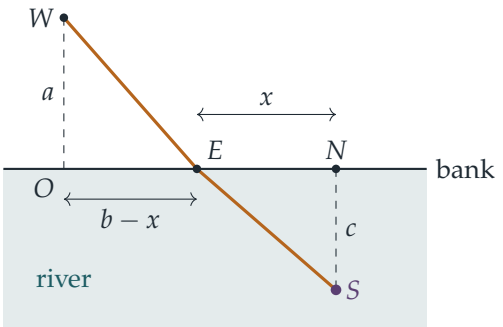
step.maths.org/questions/87-s3-q11

The key idea

Write the total time as a function of the single variable x and set its derivative to zero. In still water this is just the refraction problem from optics, and the answer is Snell's law. With a current the only new work is the swimming time: the current and her swimming velocity have the same size, so their sum is the diagonal of a rhombus, which makes the swimming time a surprisingly simple function of x.

Solution

Model the woman as a particle moving at constant speed u on land and at constant speed v relative to the water, each along a straight line. Take the bank as the X-axis with downstream as the positive direction, the field as $Y > 0$, and put the woman at $W = (0, a)$. The stump is then $S = (b, -c)$, the nearest point of the bank to it is $N = (b, 0)$, and she enters the river at $E = (b - x, 0)$, which is x upstream of N.



Her running distance is $WE = \sqrt{a^2 + (b-x)^2}$ and her swimming distance is $ES = \sqrt{x^2 + c^2}$.

Still water

The total time is

$$T(x) = \frac{\sqrt{a^2 + (b-x)^2}}{u} + \frac{\sqrt{x^2 + c^2}}{v},$$

so

$$T'(x) = -\frac{b-x}{u\sqrt{a^2 + (b-x)^2}} + \frac{x}{v\sqrt{x^2 + c^2}}.$$

The function $s \mapsto s/\sqrt{k^2 + s^2}$ is increasing, so both terms of T' increase with x. Hence T' is strictly increasing and T has exactly one stationary point, which is its minimum. Since $T'(0) < 0 < T'(b)$, this point lies in $0 < x < b$. The required equation is

$$\boxed{\frac{b-x}{u\sqrt{a^2 + (b-x)^2}} = \frac{x}{v\sqrt{x^2 + c^2}}}$$

or equivalently $v(b-x)\sqrt{x^2 + c^2} = ux\sqrt{a^2 + (b-x)^2}$. In terms of the angles α and β that WE and ES make with the normal to the bank, this reads $\dfrac{\sin\alpha}{u} = \dfrac{\sin\beta}{v}$, which is Snell's law of refraction.

Flowing river

Now the water moves downstream at speed v, and she swims at speed v relative to the water. To travel along the straight line ES she must choose a heading, at angle θ below the downstream direction, such that her velocity relative to the ground points along ES.

Ask yourself How fast does she actually move along ES, and why must she enter upstream of the stump?

Her ground velocity is the current plus her swimming velocity:

$$v\begin{pmatrix}1\\0\end{pmatrix} + v\begin{pmatrix}\cos\theta\\-\sin\theta\end{pmatrix} = 2v\cos\tfrac{\theta}{2}\begin{pmatrix}\cos\frac{\theta}{2}\\-\sin\frac{\theta}{2}\end{pmatrix},$$

using $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$ and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$. Geometrically, the two velocities have equal length, so they form a rhombus and their sum bisects the angle between them. Her ground path therefore makes angle $\phi = \theta/2$ below the downstream direction. Because $0 \le \theta \le \pi$, we have $0 \le \phi \le \frac{\pi}{2}$, so she can only drift downstream: the stump must be downstream of E, which means $x > 0$. For the path to run along ES we need

$$\tan\phi = \frac{c}{x}, \qquad \text{so} \qquad \cos\phi = \frac{x}{\sqrt{x^2 + c^2}},$$

and her speed along ES is $2v\cos\phi = \dfrac{2vx}{\sqrt{x^2 + c^2}}$. The swimming time is distance over speed:

$$\frac{\sqrt{x^2 + c^2}}{2vx/\sqrt{x^2 + c^2}} = \frac{x^2 + c^2}{2vx}.$$

So the total time is

$$T(x) = \frac{\sqrt{a^2+(b-x)^2}}{u} + \frac{x^2+c^2}{2vx} = \frac{\sqrt{a^2+(b-x)^2}}{u} + \frac{1}{2v}\left(x + \frac{c^2}{x}\right), \qquad x > 0,$$

and

$$T'(x) = -\frac{b-x}{u\sqrt{a^2+(b-x)^2}} + \frac{1}{2v}\left(1 - \frac{c^2}{x^2}\right).$$

Both terms again increase strictly with x for $x > 0$. Also $T'(x) \to -\infty$ as $x \to 0^+$, and $T'(x) > 0$ for large x, so there is exactly one stationary point and it gives the minimum time. Setting $T'(x) = 0$ and multiplying by $2vx^2u\sqrt{a^2+(b-x)^2}$ (which is positive) gives

$$2vx^2(b-x) = u(x^2-c^2)[a^2+(b-x)^2]^{\frac{1}{2}},$$

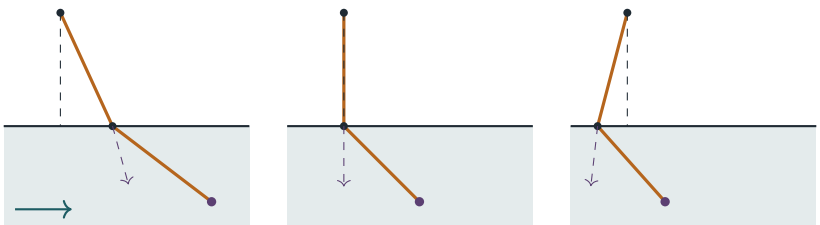
as required.

The three sketches

The equation tells us where x lies without solving it. The factor $[a^2+(b-x)^2]^{1/2}$ is positive, so $b-x$ and $x-c$ must have the same sign (or both be zero).

- $b > c$: then $c < x < b$. If $x \le c$ we would have $b - x > 0 \ge x - c$; if $x \ge b$ we would have $b - x \le 0 < x - c$; neither is possible. She enters at $b - x > 0$, downstream of O, the point of the bank nearest her. Since $x > c$, her swimming path makes an angle $\phi < 45°$ with the bank, and her heading $\theta = 2\phi < 90°$ is angled downstream.
- $b = c$: then $x = c = b$, by the same argument. She runs straight to O, heads directly across the river ($\theta = 90°$), and the current carries her along a path at $45°$ to the bank.
- $b < c$: then $b < x < c$, so $b - x < 0$. She runs *upstream* to enter the river, swims along a path steeper than $45°$, and her heading $\theta > 90°$ points partly upstream.

The sketches below are drawn to scale for $a = 3$, $c = 2$, $u = 2$, $v = 1$, with $b = 4, 2, 1$. Numerically, $x \approx 2.62$, $x = 2$ and $x \approx 1.79$. The current flows to the right; the dashed arrow at E shows the direction she heads relative to the water.



$b > c$ $\qquad$ $b = c$ $\qquad$ $b < c$

Checks

- If $b = c$, then $x = c$ makes both sides of the given equation zero, as found above.

- Straight swimming across ($\theta = 90^\circ$): she drifts v downstream for each v across, so the path is at 45°. The formula gives speed $2v\cos 45^\circ = \sqrt{2}\,v$ along a path of length $\sqrt{2}\,c$, so the time is c/v, as it obviously must be.
- The swimming time $(x^2 + c^2)/(2vx)$ is smallest at $x = c$, by AM–GM. This makes sense: if running cost nothing she would enter the river a distance c upstream of N and head straight across.
- Numerically, minimising T directly for 2000 random choices of a, b, c, u, v always gave a minimiser satisfying both equations, and lying between b and c in the second case. Stepping the swim through the current confirmed the swimming time. For example, $a = 3$, $b = 4$, $c = 2$, $u = 2$, $v = 1$ gives $x \approx 2.621$.

Common slips

- Adding the current to her speed as if she swam straight at the stump. She must aim off so that the *resultant* velocity points at the stump.
- Forgetting that only $x > 0$ is possible with the current: with equal speeds she can't make any headway upstream.
- Squaring the equation for x too early. That loses the sign information that places x between b and c.

Beyond STEP

The first answer is Snell's law, and it comes from Fermat's principle: light takes the path of least time. Choosing a whole path to minimise a time is the subject of the calculus of variations. A close cousin is Zermelo's navigation problem: steer a boat through a current that varies from place to place so as to arrive as soon as possible. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers relative velocity and moving frames carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.